Shortest path algorithm

Shortest Path algo

- It is an algorithm for finding the shortest paths between nodes in a **graph**, which may represent, for **example**, road networks.
- The **single source shortest path algorithm** is used to find minimum distance from **source** vertex to any other vertex. At first it finds those distances which have only one edge in the **path**.

Dijkstra

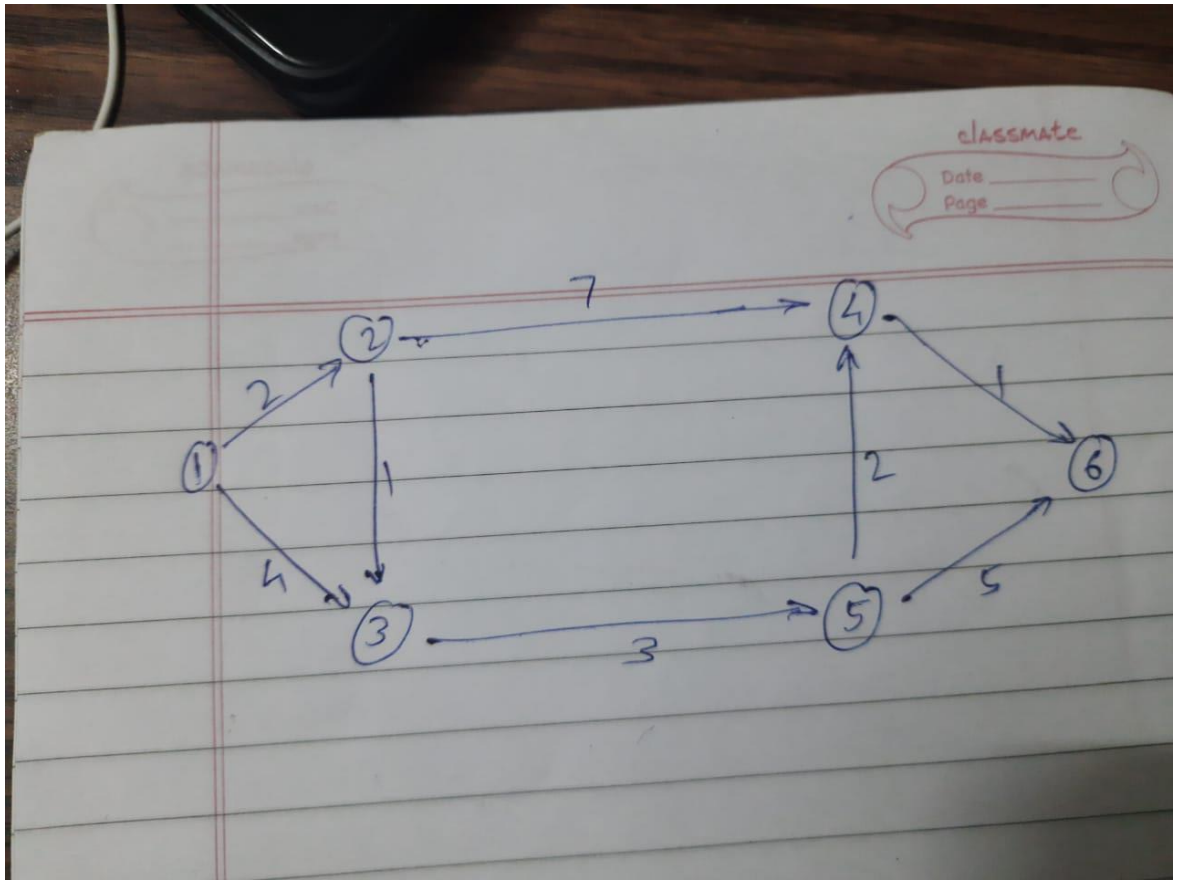
- Single source shortest path, direct or via other vertices.
- Shortest path-minimization/optimization problem-Greedy approach
- Works on directed/ undirected graph
- Relaxation

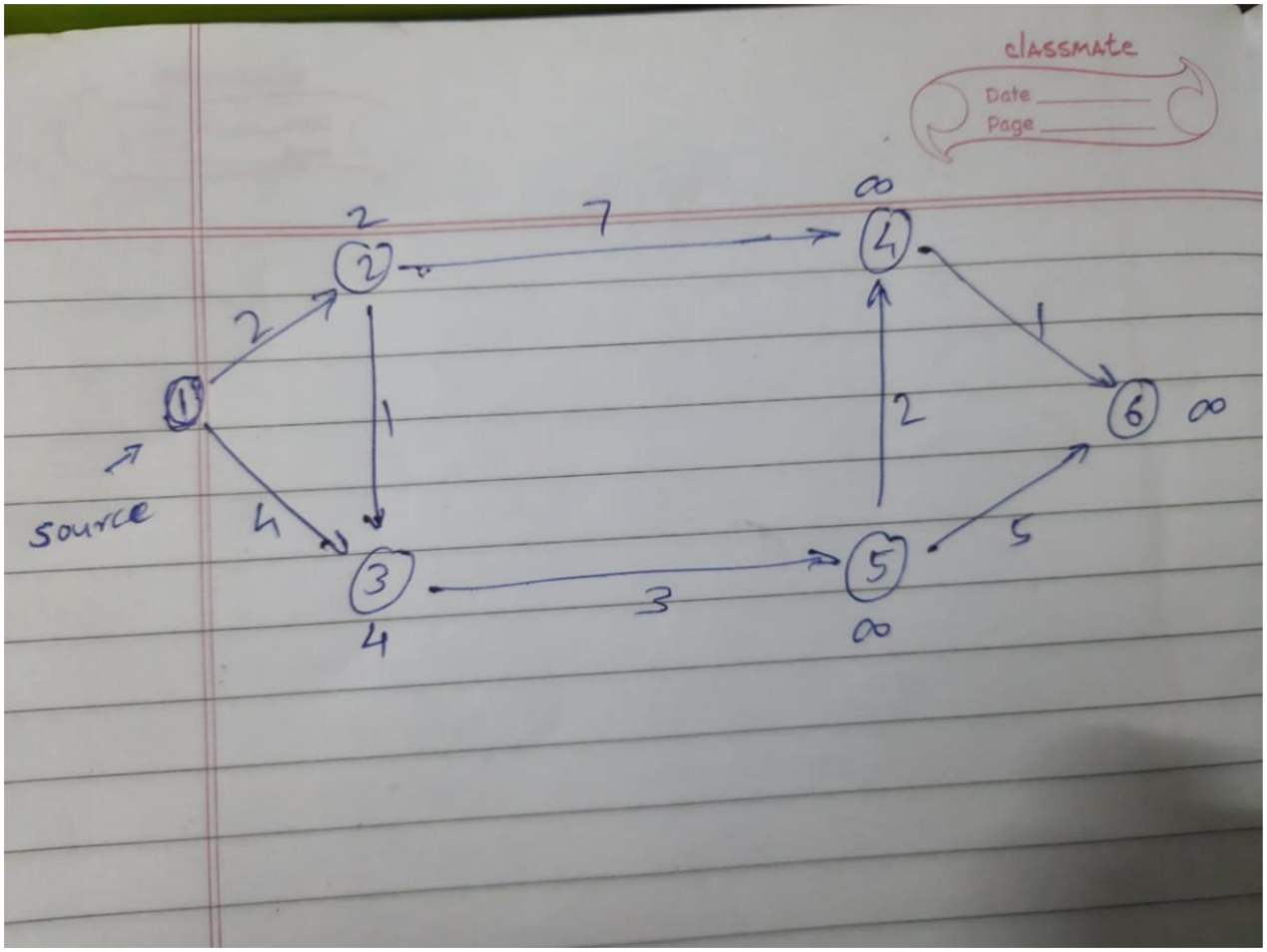
If(d[u]+c(u,v) < d[v])

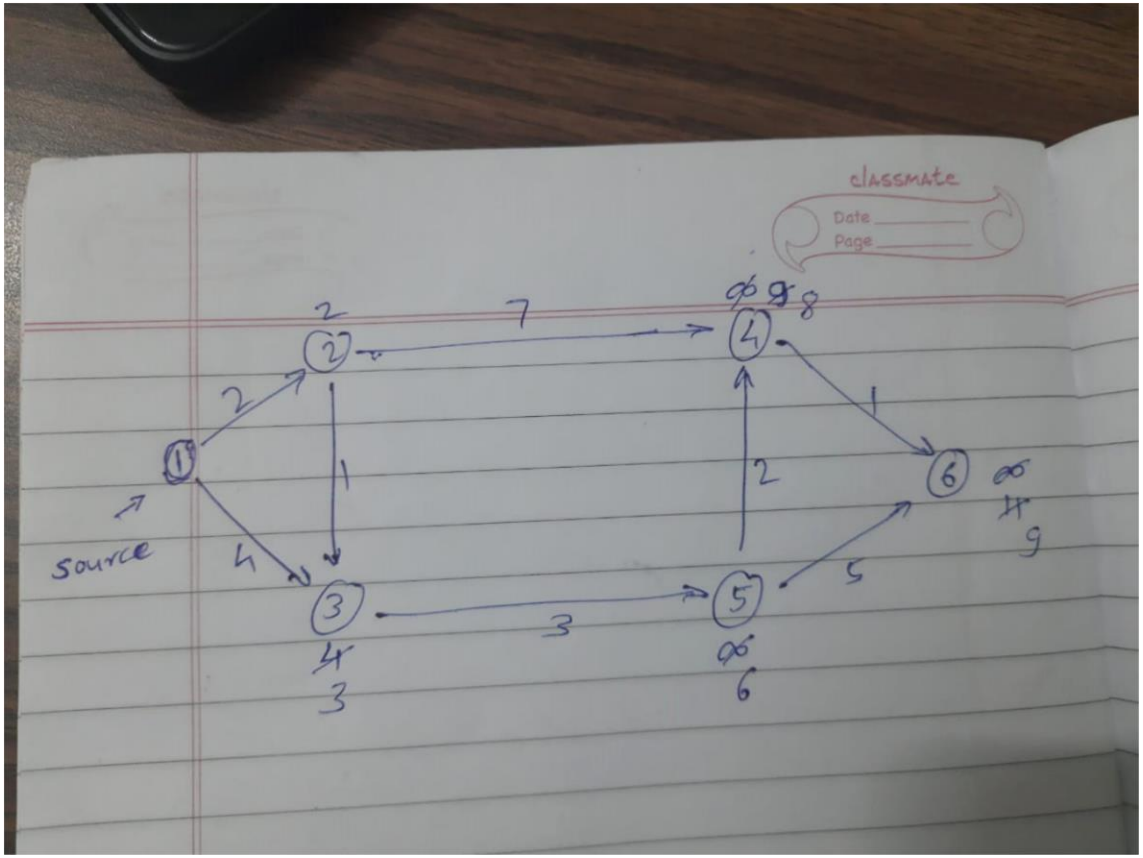
- Complexity $O(V^2)$

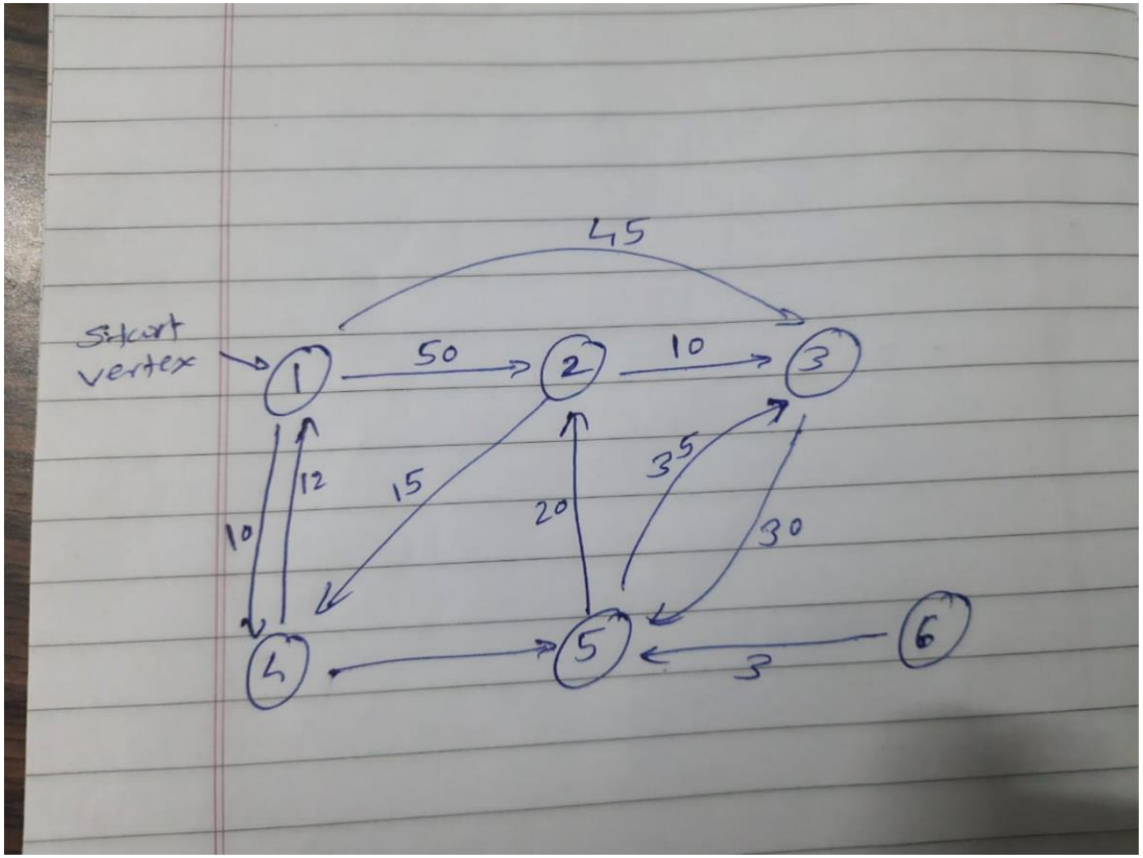
- Limitation

Cannot handle negative weights


1
2
3
4
5
6
2
4
1
7
3
2
1
5

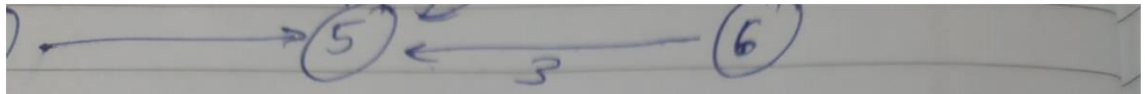

source
1
2
3
4
5
6
2
4
2
1
7
3
2
1
5
2
4
∞
∞
∞


source
1
2
3
4
5
6
2
4
7
1
3
2
5
1
2
4
3
∞ ~~9~~ 8
∞
6
∞
~~11~~
9


Start vertex
45
50
10
1
2
3
10
12
15
20
35
30
4
5
6
3

Vertex	2	3	4	5	6
1	50	45	10	∞	∞

Selected Vertex	2	3	4	5	6
	50	45	10	∞	∞
4	50	45	10	25	



Vertex	2	3	4	5	6
1	50	45	10	∞	∞
4	50	45	10	25	∞
5	45	45	10	25	∞

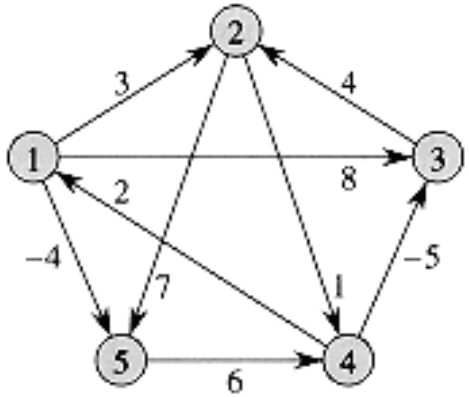
Selected Vertex	2	3	4	5	6
1	50	45	10	∞	∞
4	50	45	10	25	∞
5	45	45	10	25	∞
2	45	45	10	25	∞

Selected Vertex	2	3	4	5	6
1	50	45	10	∞	∞
4	50	45	10	25	∞
5	45	45	10	25	∞
2	45	45	10	25	∞
3	45	45	10	25	∞

Selected Vertex	2	3	4	5	6
1	50	45	10	∞	∞
4	50	45	10	25	∞
5	45	45	10	25	∞
2	45	45	10	25	∞
3	45	45	10	25	∞
6	45	45	10	25	∞

All pair shortest path

- In single source shortest path, path from source to all vertices is found
- In all pair shortest path distance of every pair of vertices are found.



$$D^{(1)} = \begin{pmatrix} 0 & 3 & 8 & \infty & -4 \\ \infty & 0 & \infty & 1 & 7 \\ \infty & 4 & 0 & \infty & \infty \\ 2 & \infty & -5 & 0 & \infty \\ \infty & \infty & \infty & 6 & 0 \end{pmatrix} \quad D^{(2)} = \begin{pmatrix} 0 & 3 & 8 & 2 & -4 \\ 3 & 0 & -4 & 1 & 7 \\ \infty & 4 & 0 & 5 & 11 \\ 2 & -1 & -5 & 0 & -2 \\ 8 & \infty & 1 & 6 & 0 \end{pmatrix}$$

$$D^{(3)} = \begin{pmatrix} 0 & 3 & -3 & 2 & -4 \\ 3 & 0 & -4 & 1 & -1 \\ 7 & 4 & 0 & 5 & 11 \\ 2 & -1 & -5 & 0 & -2 \\ 8 & 5 & 1 & 6 & 0 \end{pmatrix} \quad D^{(4)} = \begin{pmatrix} 0 & 1 & -3 & 2 & -4 \\ 3 & 0 & -4 & 1 & -1 \\ 7 & 4 & 0 & 5 & 3 \\ 2 & -1 & -5 & 0 & -2 \\ 8 & 5 & 1 & 6 & 0 \end{pmatrix}$$